

Data Centre Design

Data Centre Requirements

Data Centre Prerequisites, Required Physical Area for Equipment and Unoccupied Space, Required Power to Run All the Devices, Required Cooling and HVAC, Required Weight, Required Network Bandwidth, Budget Constraints, selecting a Geographic Location, Safe from Natural Hazards, Safe from Man-Made Disasters, Availability of Local Technical Talent, Abundant and Inexpensive Utilities Such as Power and Water, Selecting an Existing Building (Retrofitting), tier standard.

KEY CONSIDERATIONS WHEN CHOOSING A DATA CENTER

- Specific business goals
- Facility location
- Data center infrastructure
- Service reliability
- Security levels
- Required scalability
- Carrier neutrality
- On-Hand support
- Data center's reputation
- Costs and pricing
- The likelihood of disasters
- Extra amenities



Your Goals and Needs

- Upper management.
- Network admins.
- SecOps and DevOps teams.
- Software developers.
- Enhancing data center uptime.
- Improving app performance.



Required Physical Area for Equipment and Unoccupied Space

[<https://www.youtube.com/watch?v=Ds7VuIqLfRM>]

- Preferably not near residential or other sound-sensitive users
- • Maximum of one-hour distant commutation from major end users
- • Adequate site for at-grade development (for example, parking, water storage, fuel storage, transformer yard or substation or generators)
- • Ready access to power from diverse sources (such as multiple grids)
- Truck access for equipment delivery
- • Assess proximity to sources of vibration and high-risk sources (such as airports and rail lines)
- • Location outside flood plains and tornado- and hurricane-prone areas
- • Local authorities amenable to building use

Site Location

Site Location

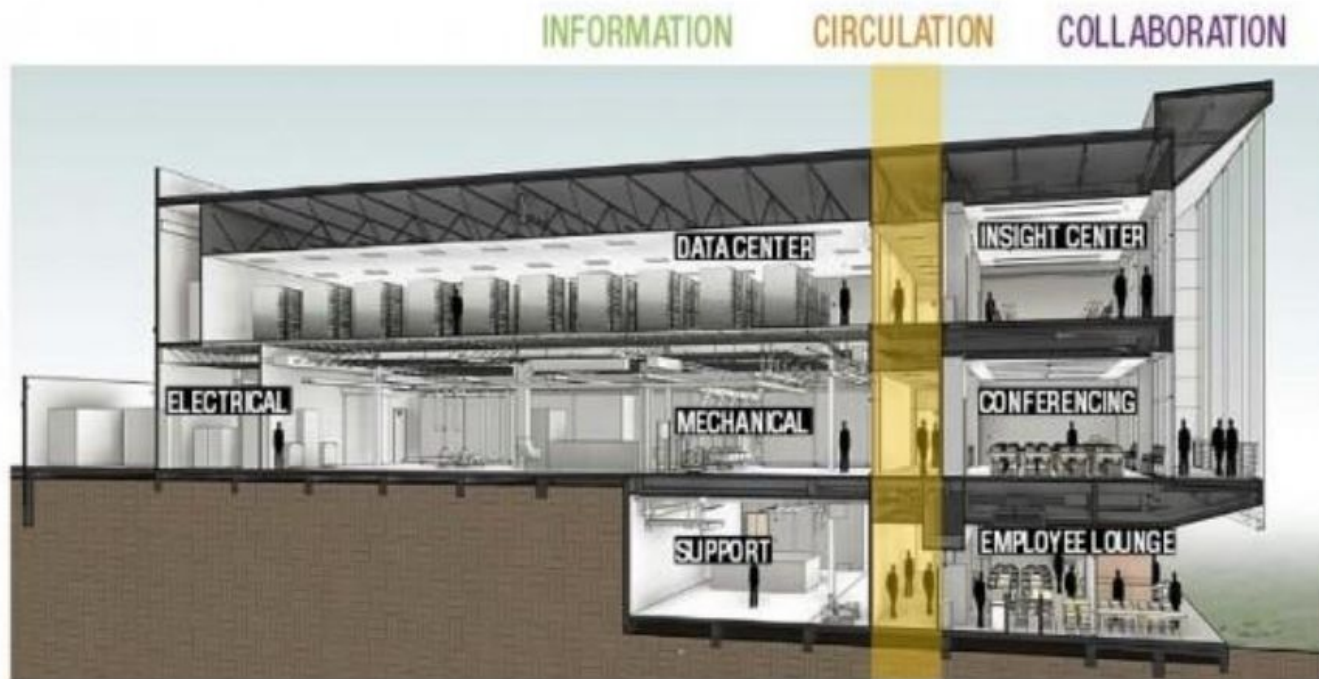


Site and Building Selection Criteria

- Larger column bays (30 feet x 50 feet is good)
Minimum 13 ½ feet clear from
 - structural slab to lowest structural member
- • Efficient floor plates (that is, rectangular, square or side core preferable)
- • 50,000 to 75,000 usable square feet preferable

Site and Building Selection Criteria

Site and Building Selection Criteria



- Loading docks for equipment delivery access
- • Compactor for rubbish removal
- • Single point of entry and sufficient setback of building for perimeter security
- purposes
- • Multiple story only: adequate riser space for primary/emergency power, HVAC,
- diverse fiber entries and other vertical services
- • Rooftop acoustic and aesthetic screening for mechanical equipment

Selecting a Geographic Location

Safe from Man-Made Disasters

1. A data center should not be located very close to an airport, electrical railways, or telecommunications signal center.
2. Avoid selecting a site too close to a mine, quarry, highway, or heavy industrial plant whose vibrations may disrupt racks and servers within the data center or utilities outside the data center.

Required Power to Run All the Devices

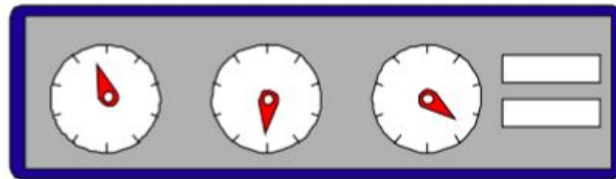


Fig. Power Distribution

Assess overall power requirements (that is, use rack unit measure).

- Strive for multiple utility feeds.
- Provide for maintenance bypass and emergency shutdown.
- Determine if equipment requires single phase or three-phase power

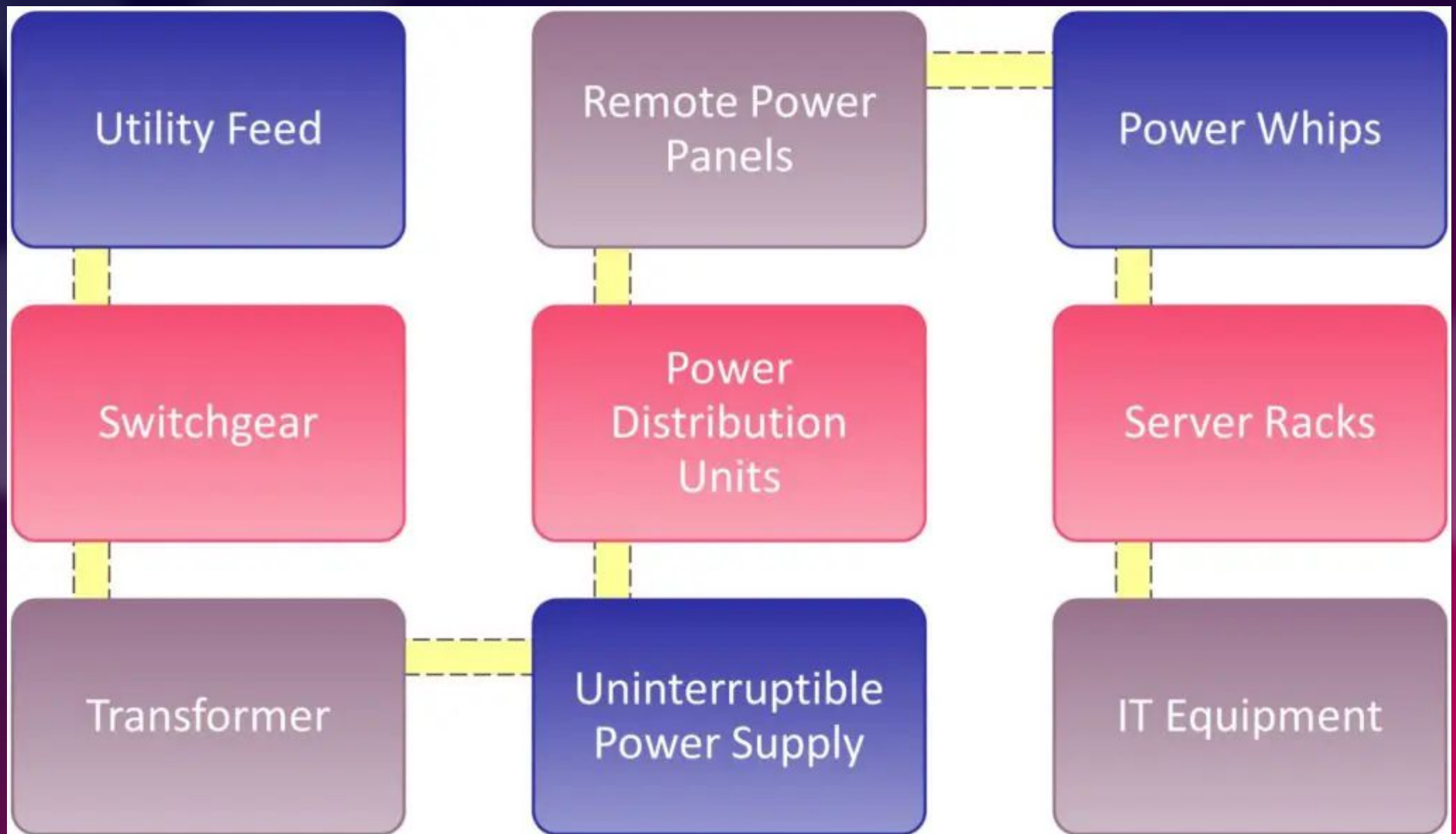
1. Why Do Data Centers Need Power?



2. How Much Power Does a Data Center Require?

Data Center Size	Small	Medium	Large
Building Size	5,000 – 20,000 sqft	20,000 – 100,000 sqft	100,000 sqft to millions of sqft
Server Count	500 – 2,000 servers	2,000 – 10,000 servers	10,000 to 100,000 servers
Power Capacity	1 – 5 MW	5 – 20 MW	20 – 100+ MW
Design/Efficiency	Basic power management and cooling	Robust power management, partial efficiency	High efficiency, renewable energy use
Example Company	Equinix	Digital Realty	Amazon Web Services

3. Data Center Power Distribution Hierarchy



Utility Power Capacity vs IT Power Capacity



Utility Power

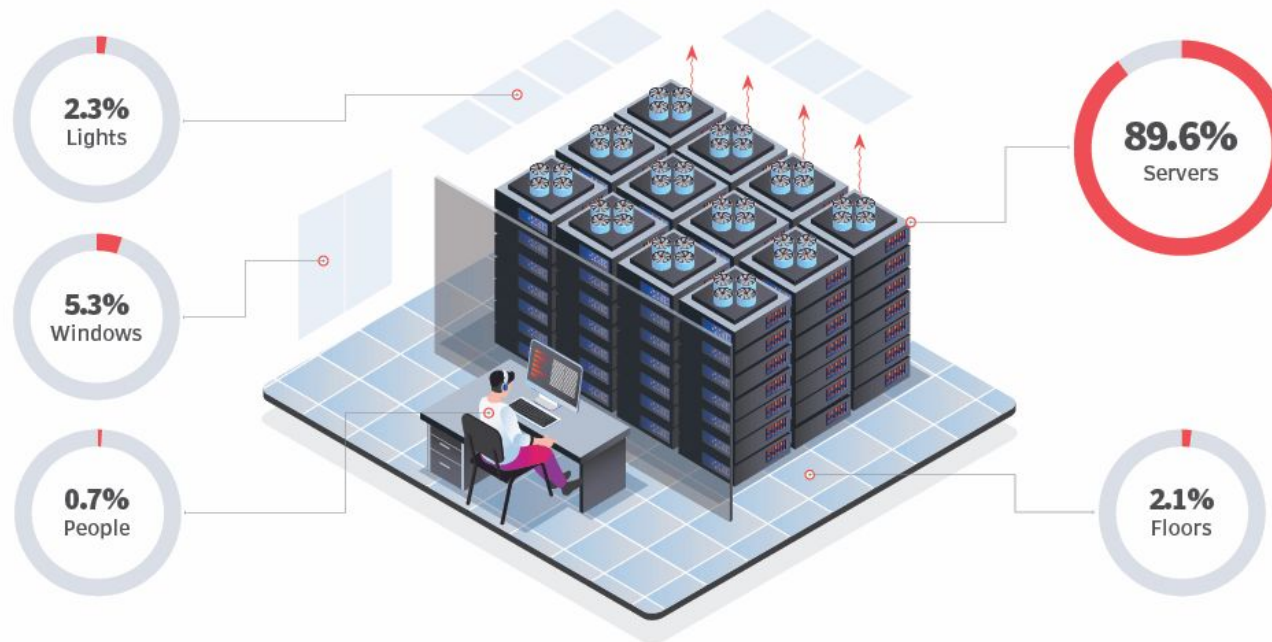


IT Power

Required Cooling and HVAC

Where's the heat?

Data center heat is generated from several sources, but primarily from the servers.

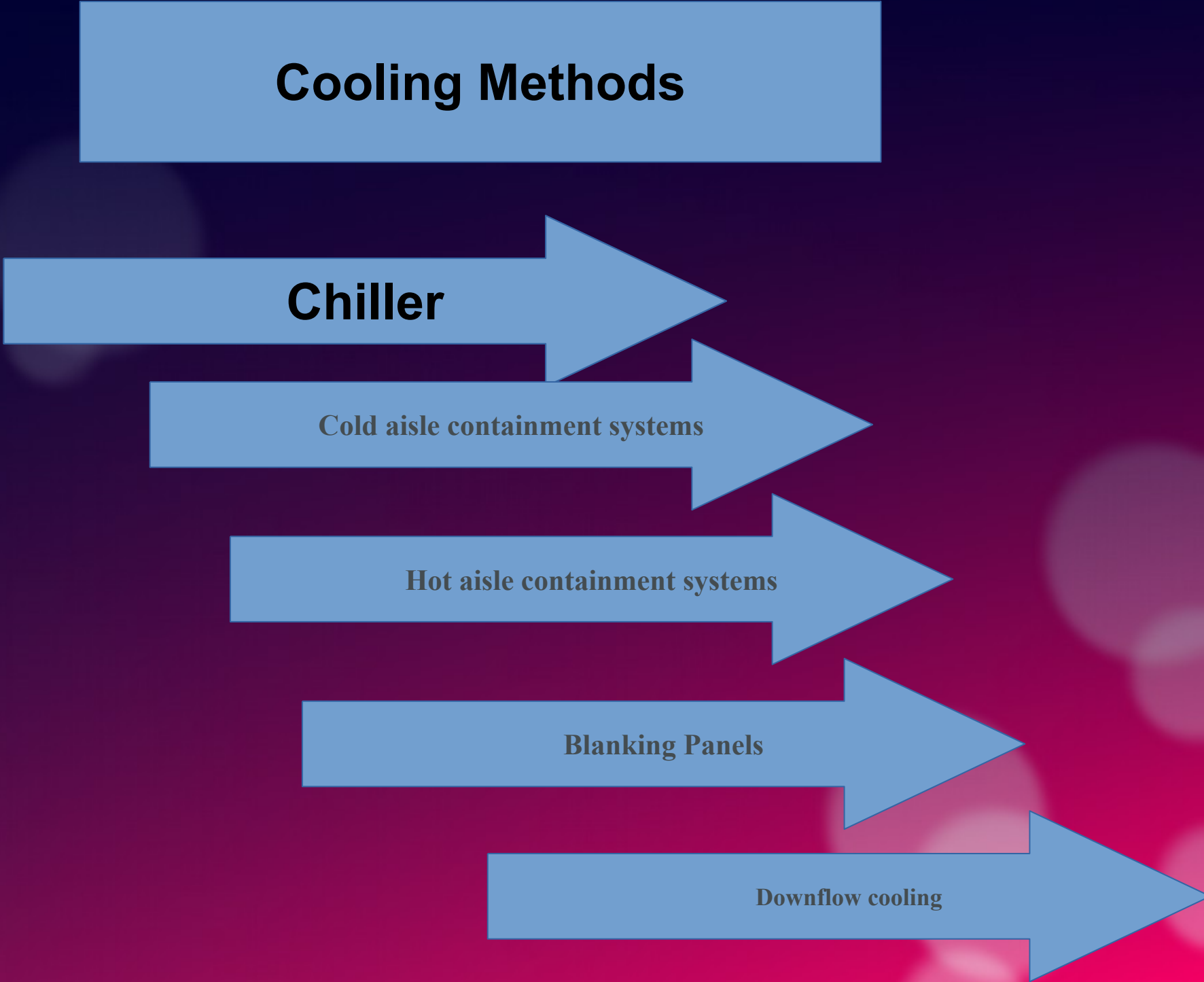


How to calculate data center cooling requirements

Data center cooling requirements are affected by several factors, including the equipment's heat output, floor area, facility design and electrical system power rating.

<https://www.youtube.com/watch?v=WtVO2NrG9Rg>

Cooling Methods



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graph TD; CM[Cooling Methods] --> Ch[Chiller]; Ch --> CACS[Cold aisle containment systems]; CACS --> HACS[Hot aisle containment systems]; HACS --> BP[Blanking Panels]; BP --> DC[Downflow cooling]
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Chiller

Cold aisle containment systems

Hot aisle containment systems

Blanking Panels

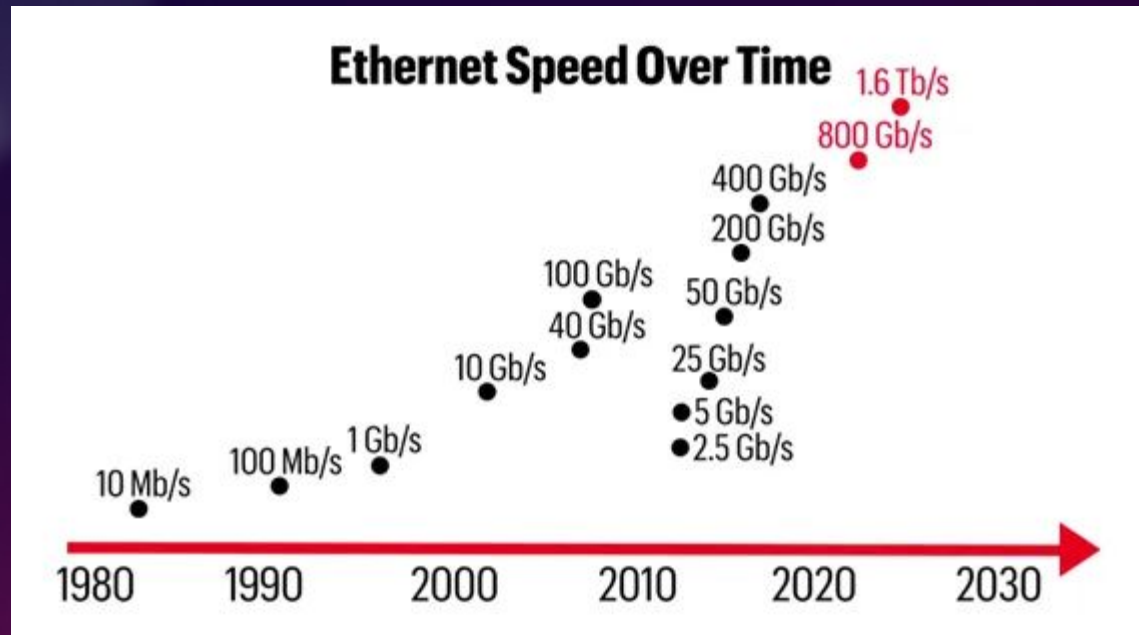
Downflow cooling

2. Calculations for a small server room or data closet In this example, there's a small server room, data closet or mini edge data center that might be found in a generic office tower in a large city.

These calculations can help determine your cooling requirements in watts, and any power system rating in kilovolt-ampere (kVA) can be roughly considered the same as the total power output of the device.

Item	Heat calculation	Output subtotal
General IT equipment: 5 servers (2,400 W each) 2 routers (18 W each)	Same as total IT power load in watts	$(5 \times 2,400) + (2 \times 18) = 12,036 \text{ W}$
UPS with battery (5 units at 0.9 kVA power system rating)	$(0.04 \times \text{power system rating}) + (0.06 \times \text{total IT load power})$	$(0.04 \times 0.9) + (0.06 \times 12,036) = 722.2 \text{ W}$
Power distribution system with 8.6 kVA	$(0.02 \times \text{power system rating}) + (0.02 \times \text{total IT load power})$	$(0.02 \times 8.6) + (0.02 \times 12,036) = 240.9 \text{ W}$
Lighting for a 10 ft x 15 ft x 10 ft room	2 x floor area (ft ²), or 21.53 x floor area (square meter)	$2 \times (10 \times 15 \times 10) = 3,000 \text{ W}$
People (150 people total in the facility)	400 x max # of people	$(400 \text{ Btu/hr} / 3.412142) \times 150 = 17,584.3 \text{ W}$
Total cooling watts needed	$12,036 + 722.2 + 240.9 + 3,000 + 17,584.3$	33,583.4 W or 33.6 kW

Network Bandwidth



Bandwidth is calculated based on total transferred data in GB

What Is Bandwidth?

Network bandwidth is a measure of the data transfer rate or capacity of a given network.

What is the 95th percentile?

The 95th percentile is the smallest number that is greater than 95% of the numbers in a given set.

The reason this statistic is so useful in measuring data throughput is that it gives a very accurate picture of data center bandwidth costs

Why Data Centers Use the 95th Percentile for Charging Bandwidth

(95th Percentile rate) x (billing cycle length) = Bandwidth usage

Example #1: Over a 30 day period, a Web server gets about 30GB of traffic.

For 28 of these days, the traffic consists of only .5GB/day.

The last two days has 8GB/day of traffic. In this case of bursting traffic, the 95th percentile would be around .5GB/day.

Using the 95th percentile calculation, the bandwidth charge would be:

Over a 30 day period, a Web server gets about 30GB of total traffic.

For 20 of these 30 days, there is only a small amount of traffic that doesn't add up at a single GB.

The remaining 10 days sees traffic of 3GB/day. The 95th percentile in this case would be 3GB.

Using the 95th percentile calculation, the bandwidth charge would be:

Budget Constraints of datacenter

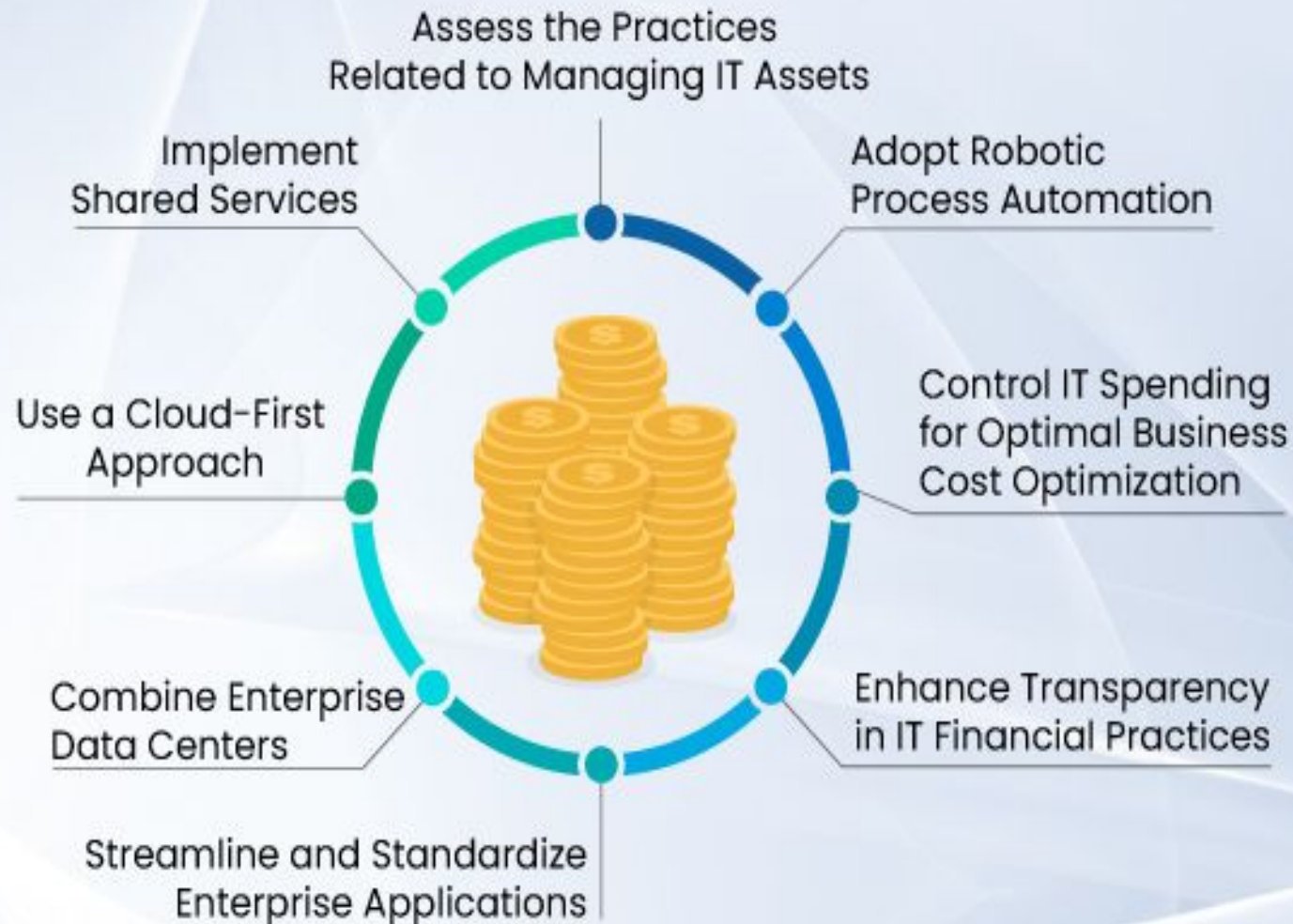
5 Main Components of Data Center Costs

- **Infrastructure and Construction costs**
 - **Servers and Storage Units**
 - **Software Licenses**
 - **Power**
 - **Network Connectivity**

How Do Data Center Costs Compare to Cloud Costs?

- 1. Capital expenditure vs. operational expenditure**
- 2. Maintenance and upgrade cost**
- 3. Energy and cooling costs**
- 4. Security and compliance costs**
- 5. Location and accessibility**
- 6. Downtime and reliability costs**
- 7. Business focus and expertise**

Techniques for Optimizing IT Costs



Selecting an Existing Building (Retrofitting)

What Is Data Center Retrofitting?

The Pros and Cons of Retrofitting an Existing Data Center

Is Retrofitting Right for You?

How to Retrofit a Data Center

What Is Data Center Retrofitting?

Data center retrofitting is the process of making significant changes to an existing data center to improve its usability.

Retrofitting presents some significant challenges

- 1.Limited flexibility
- 2.Workload disruption risk
3. Risk of high cost

Is Retrofitting Right for You?

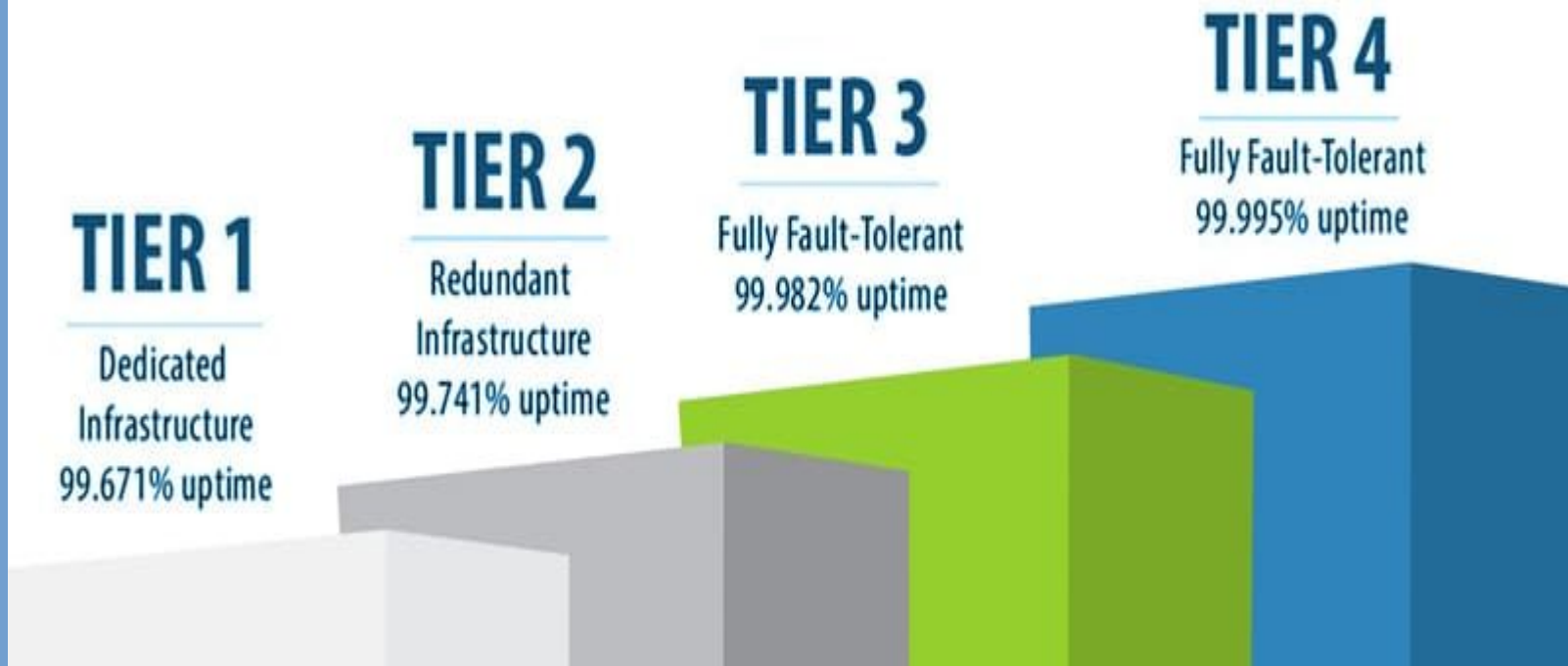
If you want to make very significant changes to your data center, you might be better off migrating to a new facility.

How to Retrofit a Data Center

1. Assess current assets
2. Determine what you want to improve
3. Establish a continuity plan
4. Test new systems
5. Migrate

Tier Standard of Data Center

DATA CENTER TIERS



1. What are the questions need to be answered while deciding the budget constraints? Discuss
2. Explain the tier standards in Data center.
3. Calculations for a small server room or data closet In this example, there's a small server room, data closet or mini edge data center that might be found in a generic office tower in a large city.
These calculations can help determine your cooling requirements in watts, and any power system rating in kilovolt-ampere (kVA) can be roughly considered the same as the total power output of the device.
4. How to Retrofit Your Data Center for Efficiency and Sustainability
5. Explain any two cooling Methods.
6. Explain the key consideration while building the data center.
7. How Do Data Center Costs Compare to Cloud Costs?
8. Explain the Data Center Power Distribution Hierarchy
9. Discuss the consideration to choose site and building criteria .
10. Whether Estimation of Network Bandwidth is necessary in Data center, Over a 30 day period, a Web server gets about 30GB of traffic. For 28 of these days, the traffic consists of only .5GB/day. The last two days has 8GB/day of traffic. In this case of bursting traffic, the 95th percentile would be around .5GB/day.
Using the 95th percentile calculation, the bandwidth charge would be